TRIMMER ASSEMBLY LINE

Monte Carlo · Yamazumi · Flow Simulation Report

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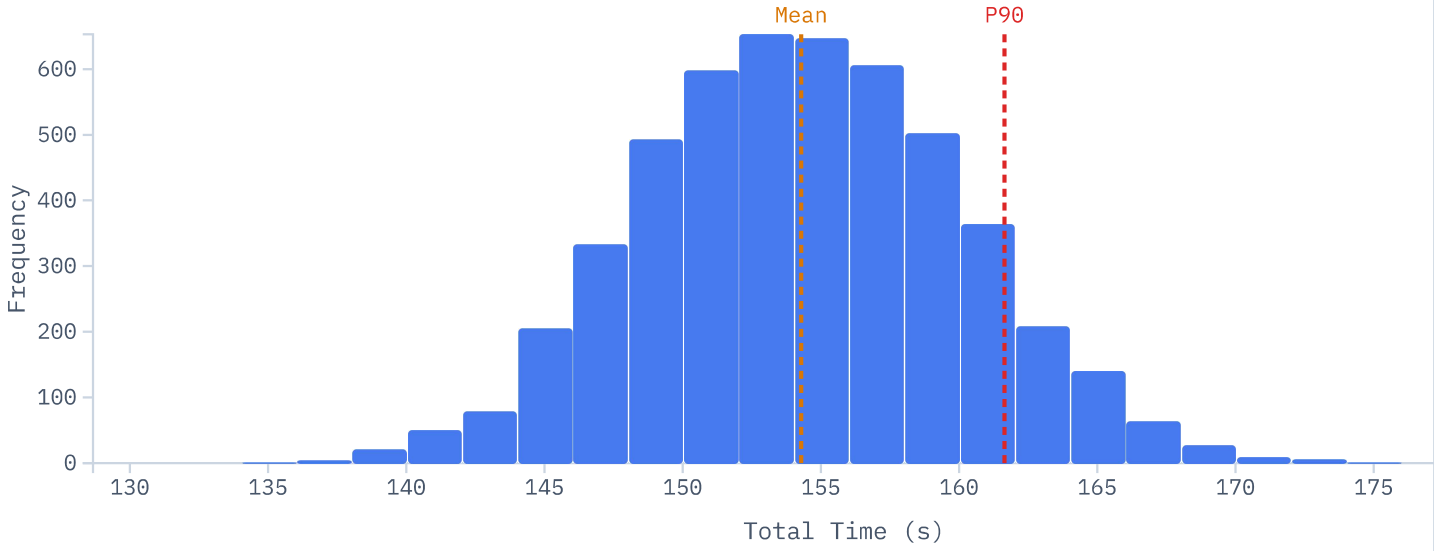
EXPECTED (MEAN)	MINIMUM
154.29 seconds	135.43 seconds

MAXIMUM	STD DEVIATION
174.32 seconds	5.73 ± seconds

DISTRIBUTION — TOTAL CYCLE TIME



Frequency Mean (154.3s) P90 (161.6s)

VALUE-ADDED ANALYSIS

VA: 141.3s (91.6%) **Semi-VA:** 10.0s (6.5%) **NVA:** 3.0s (2.0%)

SHIFT CAPACITY FORECAST & CUSTOMER DEMAND

UPDATE PLAN

AVAILABLE WORKING TIME

8 hrs 0 mins (28,800s)

REQUIRED TAKT TIME (PER UNIT)

32.00 s

EXPECTED DAILY OUTPUT (MEAN FORECAST)

951 units

P90 GUARANTEE (CONSERVATIVE)

866 units

P10 POTENTIAL (OPTIMISTIC)

1,063 units

PERCENTILE BREAKDOWN (EXPECTED OUTPUT)

PCT	OUTPUT (UNITS)	CONF.	VISUAL
P10	**866**	10%	
P25	**901**	25%	
P50	**947**	50%	
P75	**1002**	75%	
P80	**1019**	80%	
P85	**1038**	85%	
P90	**1063**	90%	
P95	**1096**	95%	
P99	**1160**	99%	

TASK CONTRIBUTION

TASK	TYPE	AVG (S)	% OF TOTAL	STN
1. tán_co_terminal	VA	26.91	17.4%	1
2. Housing_top_lò_xo	VA	12.79	8.3%	2
3.1 Positioning	SVA	4.11	2.7%	3
3.2 Tra mỡ bò lên răng dao	VA	4.67	3.0%	3
3.3 Đặt ướm dao dưới lên đầu	SVA	2.63	1.7%	3
3.4 Đặt ướm dao trên lên	SVA	3.22	2.1%	3
3.5 lấy vít	NVA	1.47	1.0%	3
3.6 siết ốc	VA	1.14	0.7%	3
4. hipot,lazer,_test_dây	VA	29.46	19.1%	4
5. xịt,_chèn,_dầu_lưỡi,_blade_guard	VA	17.95	11.6%	5
6. hdpe,_hộp_nhựa,_phụ_kiện	VA	24.12	15.6%	6
7. color_box,_keo	VA	11.53	7.5%	7
8. master_carton,_keo	VA	12.70	8.2%	8
3.7 QC khe hở dao	NVA	1.58	1.0%	3

TARGET DAILY OUTPUT ANALYSIS

PROBABILITY OF PRODUCING 900 UNITS/DAY

75.9%

≥ TARGET (MET)

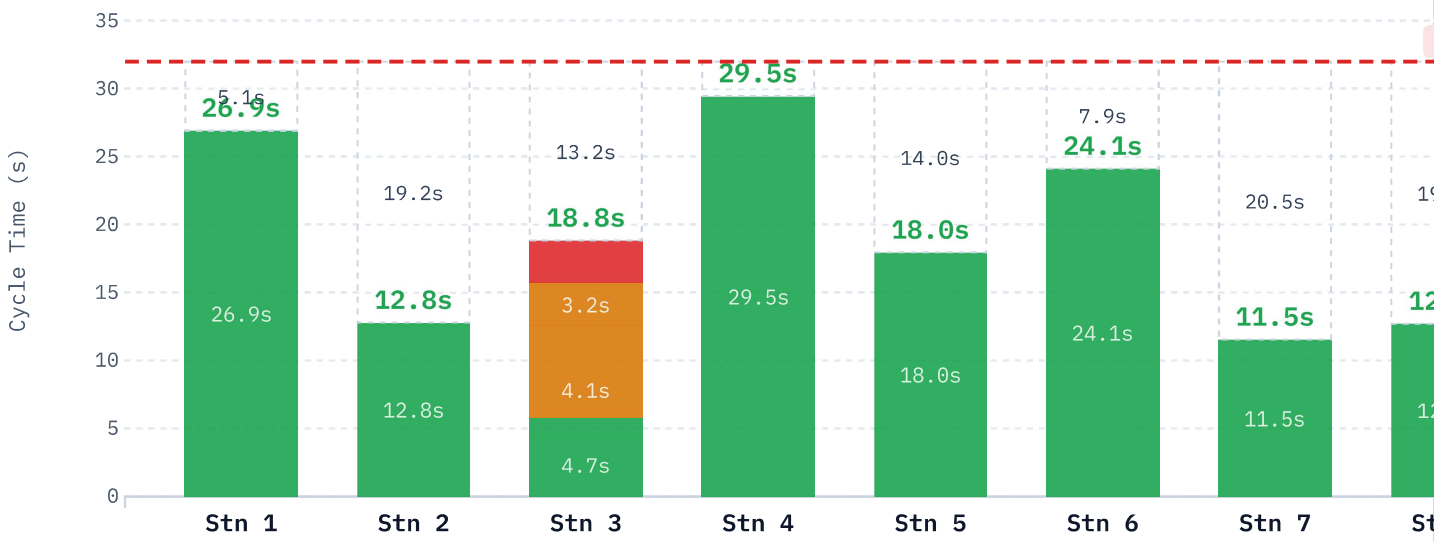
75.9%

< TARGET (MISSED)

24.1%

YAMAZUMI — OPERATOR LOADING CHART

Takt Time (s) 32.0



VA (Green) — 91.6% Semi-VA (Yellow) — 6.5% NVA (Red) — 2.0% Unbusy / Idle

STATIONS	PEAK STATION
8	Stn 4
operators	29.46s

BALANCE EFFICIENCY	OVER TAKT
65.5%	0
line balance	stations / 8

FLOW SIMULATION — BOTTLENECK & UTILIZATION ANALYSIS

500 units simulated (DES)

BOTTLENECK
Stn 4
29.67s / unit

THROUGHPUT RATE
121
units / hour

LINE EFFICIENCY
65.1%
balance rate

AVG SYSTEM WAIT
678.83s
queue delay / unit

STATION	TASKS ASSIGNED	AVG CYCLE (S)	UTILIZATION	AVG QUEUE WAIT	AVG QUEUE DEPTH	STATUS
Stn 1	1. tán_co_terminal	27.04	90.4%	0.00s	0.00	HIGH LOAD
Stn 2	2. Housing_top_lò_xo	12.67	42.4%	0.00s	0.00	OK
Stn 3	3.1 Positioning, 3.2 Tra mỡ bò lên răng dao, 3.3 Đặt ướm dao dưới lên đầu, 3.4 Đặt ướm dao trên lên, 3.5 lấy vít, 3.6 siết ốc, 3.7 QC khe hở dao	18.84	63.0%	0.01s	0.00	OK
Stn 4	4. hipot,lazer,_test_dây	29.67	99.2%	678.47s	0.98	BOTTLENECK
Stn 5	5. xịt,_chèn,_dầu_lưỡi,_blade_guard	17.94	60.0%	0.00s	0.00	OK
Stn 6	6. hdpe,_hộp_nhựa,_phụ_kiện	24.23	81.0%	0.36s	0.12	HIGH LOAD
Stn 7	7. color_box,_keo	11.47	38.3%	0.00s	0.00	OK
Stn 8	8. master_carton,_keo	12.69	42.4%	0.00s	0.00	OK